

DISTRIBUTED LOG AGGREGATION SYSTEM

COURSE: SOCKET PROGRAMMING

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AIM

To design and implement a Distributed Log Aggregation System using UDP sockets to collect logs from multiple clients and process them in real time at a central server.

PROBLEM STATEMENT

In distributed systems, logs are generated across multiple machines. Managing and analyzing these logs centrally in real time is challenging. This project aims to design a system that aggregates logs from multiple clients into a single server using UDP communication.

SYSTEM ARCHITECTURE

Client1 —▶

└─▶ Server (Log Aggregator)

Client2 —▶

ALGORITHM

◆ Server Algorithm

1. Create UDP socket
2. Bind socket to port
3. Wait for incoming messages
4. Receive message using `recvfrom()`
5. Add timestamp
6. Display log
7. Repeat

Client Algorithm

1. Create UDP socket
2. Read input message
3. Send message using `sendto()`
4. Repeat

KEY FEATURES

1. Streaming Ingestion

Logs are continuously sent from clients to the server.

2. Time Ordering

Server adds timestamps to each log.

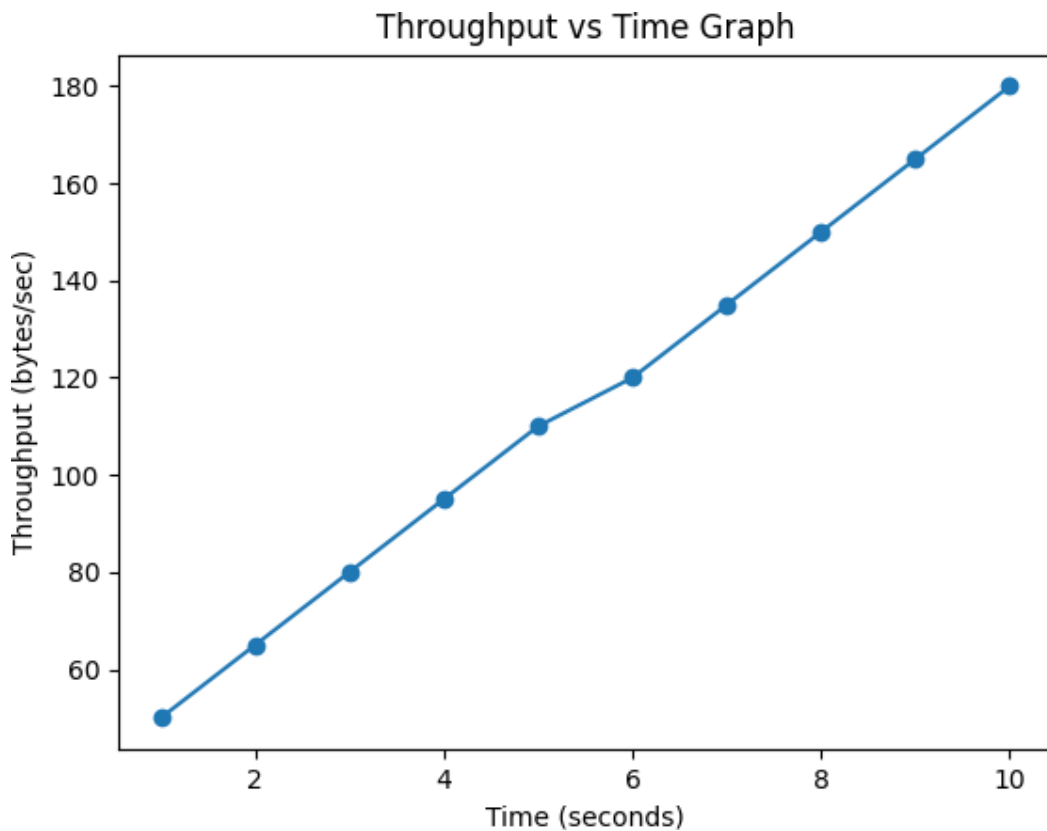
3. Backpressure Handling

Server processes logs sequentially. UDP may drop packets if overloaded.

4. Throughput Evaluation

Number of logs processed per second is observed.

GRAPH



RESULT

The system successfully collected and displayed logs from multiple clients in real time using UDP communication

CONCLUSION

This project demonstrates real-time communication and distributed log aggregation using UDP sockets. It provides an efficient and lightweight method for collecting logs from multiple sources.